

Science Communication Product Rubric

AiSciComm Module 4

Product Task Description:	Evaluate student science communication products explaining climate mechanisms, impacts, and solutions to a targeted audience.				
Science Communication Product Information:	Group/ Student(s): [GROUP #/ NAME(S)] Target Audience: _____ Science Communication Product: • Infographic • Community Event • Social Media Post (e.g., Instagram post or story) • Town Hall Presentation • Podcast • Game • Other: _____ Total Points: ____ / 24				
Component	Sub-component	Description	Meets Expectations (3 points)	Approaching (2 points)	Needs Improvement (1 point)
Clarity and Language (/ 3)	Accessibility (/ 3)	Delivers information using accessible, audience-appropriate vocabulary (low jargon).	Uses clear, accessible language throughout; explicitly defines necessary technical terms in context.	Uses mostly accessible language but includes some unexplained technical terms or jargon.	Overuses heavy jargon without definitions, making content difficult to understand for the audience.

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			<i>Example: “Ocean acidification means seawater becomes more acidic as it absorbs excess carbon dioxide, making it harder for shellfish to build their shells.”</i>	<i>Example: “Ocean acidification happens when CO2 lowers seawater pH levels, affecting marine life.”</i>	<i>Example: “Anthropogenic pCO2 flux lowers oceanic pH via carbonic acid dissociation, inhibiting calcification.”</i>
	Accessibility (OPTIONAL) (+1)	Translates information into different languages based on audiences.	Provides translations.		Does not provide translation.
Content (/ 9)	Science Content (/ 3)	Includes accuracy and depth of scientific concepts.	Clearly communicates all science ideas that are rooted in credible data or evidence. <i>Example: Defines key concepts accurately, such as explaining that ocean acidification is the lowering of seawater pH when carbon dioxide dissolves into the ocean, and that thermal expansion is ocean water taking up more space as it warms.</i>	Explains a science idea, but the explanation is difficult to understand or is missing credible data or evidence. <i>Example: Accurately defines carbon dioxide emissions and ocean warming but incorrectly defines sea level rise as being caused primarily by melting floating Arctic sea ice rather than land ice.</i>	Does not explain scientific concepts or does not explain the scientific concepts correctly. <i>Example: Defines sea level rise incorrectly by claiming that using too much tap water in homes drains into the ocean and fills it up.</i>
	Connection to Everyday Life and Society Impact	Connects climate change and impacts to their community	Explicitly links climate change to societal consequences.	Mentions real-world impacts generically.	Fails to connect science to real-world impacts.

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	(/ 3)	through human infrastructure, policy, lifestyle choices, etc.	<i>Example: As California's coastal waters acidify, juvenile Dungeness crabs suffer shell dissolution, threatening the local fishing industry and traditional local holiday crab feasts.</i>	<i>Example: Ocean acidification is bad because it is hard for crabs to grow strong shells.</i>	<i>Example: Focuses on pH statistics without referencing living systems or local human communities.</i>
	Personal Connection (/ 3)	Integrates personal experiences, local community content, or emotional resonance.	Integrates authentic personal or local connection. This could be related to community interviews/ values or personal connections. <i>Example: "Living near Long Beach, I witness king tides flooding local roads, which made me realize sea level rise isn't just a future problem."</i>	Includes personal/ local connections that feel forced or superficial. <i>Example: "I like swimming in the ocean, so climate change makes me sad."</i>	Does not include a personal or local connection. <i>Example: Provides global statistics.</i>
Knowledge Organization (/ 3)	Structure (/ 3)	Frames content logically (e.g., cause → impact → solution)	Includes a clear framing and progression of content. <i>Example: Structured clearly around three parts of CO2 sources (cause), then ocean absorption (impact), and finally local</i>	Includes some structure, but transitions rapidly between topics. <i>Example: Structured clearly around three parts of CO2 sources (cause), then local mitigation steps (solution), and finally</i>	Does not have a logical structure when sharing information. <i>Example: Random facts about polar bears, then shares chemical formulas, and finally discusses climate policies</i>

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			<i>mitigation steps (solution).</i>	<i>ocean absorption (impact).</i>	<i>without a coherent structure.</i>
Style (/ 3)	Stylistic Components (/ 3)	Uses storytelling, metaphors, or engaging visuals to illustrate climate issues.	Uses narrative or strong visual imagery. <i>Example: References with uses of images to a recent event in California and the impact it has on climate issues.</i>	Uses a visual or narrative element that is weak, distracting, or unrelated. <i>Example: Includes a picture of melting glaciers but the picture does not match the text being discussed.</i>	Does not include stylistic components. <i>Example: Uses only words, no visuals.</i>
Audience Engagement (/ 6)	Audience Perspective (/ 3)	Tailors messaging specifically to the target audience (e.g., high school peers, Orange County/ Los Angeles community)	Clearly defines an audience and intentionally crafts tone and delivery for the target audience. <i>Example: Designed as an action-oriented school poster emphasizing youth climate initiatives.</i>	Selects an audience but inconsistently crafts tone and delivery for the target audience. <i>Example: Uses childish cartoon graphics alongside high-level public data for a town hall presentation designed for state and city representatives.</i>	Does not seem to have chosen a specific audience and/ or misjudges the audience background completely. Uses inappropriate tone and delivery. <i>Example: Uses overly dense scientific policy jargon and images in a board game for younger students.</i>
	Dialogic Exchange (/ 3)	Invites audience engagement, reflection, or civic action.	Prompts specific reflection or community action from the audience.	Provides a generic call to action.	Does not invite audience engagement, reflection, or civic action.

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			<i>Example: Comment on social media post "What is one way your household or school can reduce fossil fuel consumption this week?"</i>	<i>Example: Infographic sharing "Remember to reduce your carbon footprint!"</i>	<i>Example: Town hall presentation ends without any call to action, question, or summary prompt.</i>
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